

TEST PLAN

Audit Record Missing for Uncoding Dictionary Values

Field	Detail
JIRA Story	
Application	
Branch	
Prepared By	Jeevan Paapagaari, Senior QA Engineer
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Classification	CONFIDENTIAL — Internal QA Use Only
Total Test Cases	18
Priority Coverage	P1-Critical: 4 P2-High: 8 P3-Medium: 4 P4-Low: 2
Study Details	

SCOPE OF THIS TEST PLAN

This plan validates the complete audit trail functionality for the EDC-42360 fix: audit records must be written to AuditValueChange_tbl whenever a datum is uncoded. It covers 18 test cases across 7 categories: Core Functionality, User Identity & Permissions, Bulk Operations, Data Variations, Audit Trail Completeness, Performance, and Regression. Zero production leakage tolerance is enforced at P1 and P2 levels.

1. Objective

The primary objective of this test plan is to validate that the Medrio application correctly generates audit records in AuditValueChange_tbl whenever a datum is uncoded. The fix (EDC-42360) introduced a new resource DatumUncodingToString and a new CollectionName='RemoveCoding'. This plan ensures:

- A RemoveCoding row is inserted into AuditValueChange_tbl on every uncode action.
- CollectionName is exactly 'RemoveCoding' — no variation.
- TargetObjectDescription uses DatumUncodingToString format: 'Term {term_name}: Uncoded By {user} at {date}'.
- Behavior is consistent across single and multiple data points, different users/roles, different dictionaries/terms, and bulk uncoding.
- No regression introduced to coding audit behavior, permissions, other audit modules, or general audit logging.
- Zero production leakage: no uncoding action can go unaudited post-release.

2. Test Scope

2.1 In Scope

#	Area	Description
1	Uncoding Audit Generation	RemoveCoding entries in AuditValueChange_tbl
2	Coding Audit (Regression)	Verify ApplyCoding/DatumCodingToString unchanged
3	DB Backend Validation	Direct SQL queries to AuditValueChange_tbl
4	User Attribution	Correct user in audit entry (single/multi-user)
5	Bulk Uncoding	Per-datum audit rows for batch operations
6	Data Variations	Long text, special characters, multiple dictionaries
7	Date/Time Integrity	Timestamp consistency in description vs. DB column
8	Permissions	Unauthorized user cannot trigger audit entry
9	Mixed Audit Events	RemoveCoding co-exists with other audit types
10	Performance (If required)	System behavior under 100+ uncode operations
11	Concurrency	Two simultaneous users uncoding different datums
12	Network Failure	No partial audit rows on interrupted request
13	Audit UI Display	RemoveCoding entries visible in audit trail screen

2.2 Out of Scope

- Deep localization / i18n of DatumUncodingToString resource (English-only by design).
- Non-coding-related audit tables and modules beyond AuditValueChange_tbl.
- UI visual/UX design of the coding interface.
- Full load testing / performance benchmarking beyond basic stress check.
- Production DB validation (QA environment only).

3. Test Environment

Component	Specification	Notes
Application Build	MedrioEDC-1.0.0.701.exe (or later containing EDC-42360 fix)	Confirm build number before execution
Branch	dev-bran-07	Verify via app version footer or build manifest
Test URL	https://dev-bran-07-identity.medrio.org/identity/	QA environment — not production

Component	Specification	Notes
Database	AuditValueChange_tbl — direct query access required (Read)	DBA approval is needed if access pending
Dictionary	MedDRA and/or WHODrug — pre-loaded	Minimum 20 terms; backup dictionary available
Browser	Chrome latest + Firefox (regression verification)	Clear cache before each session
DB Tool	SQL Server Management Studio or equivalent	Read-only SELECT permissions minimum

4. Test Data & Users

4.1 User Accounts

User	Role	Permissions	Purpose
jp_coder01 / Medrio@123	Clinical Coder	Code, Uncode, View Audit	Primary actor for most test cases
jp_coder02 / Medrio@123	Clinical Coder	Code, Uncode, View Audit	Cross-user attribution tests (TC-003, TC-015)
jp_readonly / Medrio@123	Read-Only / Viewer	View Only — no code/uncode	Permission boundary tests (TC-006)

4.2 Test Data Requirements

Data Type	Requirement	Owner
Study	1 study with MedDRA + WHODrug dictionaries configured	QA Engineer
Form	'cdefrm' form with at least 1 codable text variable	QA Engineer
Subjects	Minimum 10 subjects (EDC-42360-001 to EDC-42360-010) with form instances	QA Engineer
Coded Terms (Short)	'Headache', 'Nausea', 'Fatigue', 'Pyrexia', 'Chest Pain'	QA Engineer
Coded Terms (Long)	Term text > 200 characters (create if not in dictionary)	QA Engineer / Data Team
Coded Terms (Special)	Non-ASCII: e.g., 'Nausée', 'Anti-TNF-α', or CJK term	QA Engineer / Data Team
Bulk Data (Optional)	100+ coded data points for performance test (TC-014)	QA Engineer
DB Baseline	Row counts recorded before each test case	QA Engineer

5. Entry & Exit Criteria

5.1 Entry Criteria

- Build MedrioEDC-1.0.0.701.exe (or later) deployed on branch dev-bran-07 and verified stable.
- QA test environment fully accessible (URL confirmed reachable).
- DB access credentials to AuditValueChange_tbl confirmed and tested.
- All 3 test user accounts were created with correct permissions.
- Study, form, and dictionary configuration completed and verified.
- Test data (subjects, coded terms) prepared and baseline row counts documented.
- EDC-42360 JIRA marked 'FB - Ready for Testing' by Development Team.
- No P1 environment blockers outstanding.

5.2 Exit Criteria

- All 18 test cases were executed and results documented.
- 100% of P1-Critical test cases (TC-001 to TC-004) passed.
- Zero P1-Critical defects open at release gate.
- Zero P2-High defects open without PM-approved deferral.
- AuditValueChange_tbl RemoveCoding entries validated via DB query for all P1 cases.

- Regression suite (TC-011, TC-012, TC-018) fully passed.
- Test execution report and defect log completed.

6. Risks & Mitigations

Risk	Severity	Likelihood	Mitigation	Owner
DB access unavailable	P1-Critical	Medium	1. Escalate to DBA team before start validation 2. Fallback to UI-only audit log review	QA Engineer
Bulk uncode feature not yet implemented	P2-High	Medium	1. TC-006 and TC-007 marked N/A 2. Document as known gap for next sprint	QA Engineer
Environment instability / build rollback	P2-High	Low	1. Daily environment health check 2. Coordinate with DevOps for build freeze	DevOps
Special character terms unavailable in dictionary	P3-Medium	Medium	1. Manually inject term into test dictionary or mock at form level	Data Team
Ambiguity in DatumUncodingToString format spec	P3-Medium	Low	1. Clarify with Dev team before test begins 2. Use existing coding entry as format reference	QA Engineer
Concurrent test interference between testers	P3-Medium	Medium	1. Assign distinct subject ranges per tester 2. Use isolated study instances	QA Lead

7. Test Case Summary

TC ID	Title	Category	Priority	Status
TC-001	Basic Single Datum Coding then Uncoding — Audit Verification	Core Functionality	P1-Critical	Not Run
TC-002	Multiple Datum Points — Individual Uncoding Audit Isolation	Core Functionality	P1-Critical	Not Run
TC-003	Uncoding by Different User — Correct User Attribution in Audit	User Identity & Permissions	P1-Critical	Not Run
TC-004	Repeated Code/Unicode Cycles — Full Ordered Audit Trail	Audit Trail Completeness	P1-Critical	Not Run
TC-005	Attempt to Uncode an Already-Uncoded Datum — No Duplicate Audit	Boundary & Negative	P2-High	Not Run
TC-006	Unauthorized User Attempt to Uncode — Audit Must NOT Be Created	Boundary & Negative	P2-High	Not Run
TC-007	Bulk Uncoding — One Audit Row Per Datum, No Aggregation (Optional)	Bulk Operations	P2-High	Not Run
TC-008	Long Term Name — Audit Description Storage Without Truncation or Error	Data Variations	P2-High	Not Run
TC-009	Special & Non-ASCII Characters in Term Name — Encoding Integrity	Data Variations	P2-High	Not Run
TC-010	Date/Time Accuracy — TargetObjectDescription Timestamp Matches DB Timestamp	Audit Trail Completeness	P2-High	Not Run
TC-011	Regression — Coding-Only Scenario (No Uncode) — ApplyCoding Audit Unchanged	Regression	P2-High	Not Run
TC-012	Mixed Operations — Audit Completeness with Data Entry, Code, Uncode, Edit	Regression	P2-High	Not Run

TC ID	Title	Category	Priority	Status
TC-013	Multiple Dictionaries — Audit Correct Across Different Dictionary Sources	Data Variations	P3-Medium	Not Run
TC-014	Performance & Stress — Bulk Uncoding of 100+ Datums Without Timeout or Partial Audit	Performance	P3-Medium	Not Run
TC-015	Concurrent Uncode Operations by Two Users — Audit Integrity Under Concurrency	Boundary & Negative	P3-Medium	Not Run
TC-016	Network Interruption During Uncode — Audit Consistency (No Partial Write)	Boundary & Negative	P3-Medium	Not Run
TC-017	Audit Trail UI Display — RemoveCoding Entries Visible in Audit Log Screen	UI Validation	P4-Low	Not Run
TC-018	Regression — Other Audit Modules Unaffected (Non-Coding Audit Flows)	Regression	P4-Low	Not Run

8. Detailed Test Cases

Each test case includes Pre-conditions, Step-by-step actions, Expected results, DB validation queries, Pass criteria, and Execution tracking fields. You can access it by using the link [Open Test Cases File](#)

09. Traceability Matrix

TC ID	Requirement / Acceptance Criteria	DB Table	Priority
TC-001	RemoveCoding row inserted on single uncode	AuditValueChange_tbl	P1-Critical
TC-002	One row per datum for multiple uncodes	AuditValueChange_tbl	P1-Critical
TC-003	UserName in audit = uncoding user identity	AuditValueChange_tbl	P1-Critical
TC-004	Complete ordered audit trail across cycles	AuditValueChange_tbl	P1-Critical
TC-005	No duplicate rows for already-uncoded datum	AuditValueChange_tbl	P2-High
TC-006	Unauthorized user cannot create RemoveCoding entry	AuditValueChange_tbl + User Permissions	P2-High
TC-007	Bulk uncode = N rows (not 1 aggregated row)	AuditValueChange_tbl	P2-High
TC-008	Long term name stored without truncation	AuditValueChange_tbl	P2-High
TC-009	Special characters stored correctly (encoding)	AuditValueChange_tbl	P2-High
TC-010	Date in description = AuditDate column value	AuditValueChange_tbl	P2-High
TC-011	Regression: ApplyCoding rows unchanged	AuditValueChange_tbl	P2-High
TC-012	Mixed events — full lifecycle audit completeness	AuditValueChange_tbl	P2-High
TC-013	Multi-dictionary consistency	AuditValueChange_tbl	P3-Medium
TC-014	Performance: 100+ uncodes, no timeout/partial write	AuditValueChange_tbl	P3-Medium
TC-015	Concurrency: 2 users, correct attribution	AuditValueChange_tbl	P3-Medium
TC-016	Network failure: no partial audit row	AuditValueChange_tbl	P3-Medium
TC-017	UI audit trail displays RemoveCoding entries	UI Audit Log	P4-Low
TC-018	Regression: non-coding audits unaffected	AuditValueChange_tbl	P4-Low

10. Defect Management

10.1 Severity Classification

Severity	Definition	Example in EDC-42360 Context	Response SLA
S1 — Critical	1. System cannot be used 2. Data loss 3. Regulatory non-compliance	1. RemoveCoding row not created at all (bug not fixed)	Immediate
S2 — High	1. Major functionality broken. 2. Workaround difficult	1. Wrong user attributed in audit. 2. Timestamp off by 24h. 3. duplicate rows	Within 1 business day
S3 — Medium	1. Functionality impaired but workaround exists	1. Special character stored with minor corruption. 2. UI doesn't display entry	Within 2 business days
S4 — Low	1. Minor cosmetic or edge-case issue	1. Timestamp format slightly different from other audit entries	Within 1 sprint

10.2 Defect Log Template

Defect ID	TC ID	Title	Severity	Steps to Reproduce	Expected	Actual	Status	Assigned To
							Open	

10.3 Release Gate

RELEASE GATE — ZERO TOLERANCE POLICY

This build is BLOCKED from production release if ANY of the following are true:

- Any P1-Critical test case (TC-001 to TC-004) fails and any S1 or S2 defect is open without Product Owner deferral sign-off.
- DB validation (AuditValueChange_tbl) confirms RemoveCoding rows are missing for any uncode operation.
- Regression test cases TC-011, TC-012, or TC-018 fail.

11. Test Execution Summary Tracker

Complete this section upon finishing test execution. Update before review meeting.

TC ID	Priority	Executed By	Execution Date	Result	Defect ID
TC-001	P1-Critical			Pass / Fail / Blocked / N/A	
TC-002	P1-Critical			Pass / Fail / Blocked / N/A	
TC-003	P1-Critical			Pass / Fail / Blocked / N/A	
TC-004	P1-Critical			Pass / Fail / Blocked / N/A	
TC-005	P2-High			Pass / Fail / Blocked / N/A	
TC-006	P2-High			Pass / Fail / Blocked / N/A	
TC-007	P2-High			Pass / Fail / Blocked / N/A	
TC-008	P2-High			Pass / Fail / Blocked / N/A	
TC-009	P2-High			Pass / Fail / Blocked / N/A	
TC-010	P2-High			Pass / Fail / Blocked / N/A	

TC ID	Priority	Executed By	Execution Date	Result	Defect ID
TC-011	P2-High			Pass / Fail / Blocked / N/A	
TC-012	P2-High			Pass / Fail / Blocked / N/A	
TC-013	P3-Medium			Pass / Fail / Blocked / N/A	
TC-014	P3-Medium			Pass / Fail / Blocked / N/A	
TC-015	P3-Medium			Pass / Fail / Blocked / N/A	
TC-016	P3-Medium			Pass / Fail / Blocked / N/A	
TC-017	P4-Low			Pass / Fail / Blocked / N/A	
TC-018	P4-Low			Pass / Fail / Blocked / N/A	

Metric	Target	Actual
Total Test Cases	18	
Executed	18 (100%)	
Passed	18 (100%)	
Failed	0	
Blocked	0	
P1-Critical Passed	4/4 (100%)	
S1 Defects Open	0	
S2 Defects Open	0 (or deferred with PO sign-off)	
Overall Status	PASS / FAIL / BLOCKED	

12. Assumptions & Dependencies

ID	Assumption / Dependency	Impact if Invalid	Owner	Status
A-01	AuditValueChange_tbl is the authoritative source for coding/uncoding audit history	All DB validation steps invalid	Dev Lead	Confirmed
A-02	CollectionName='RemoveCoding' is the correct and only expected value	TC-001 to TC-013 pass criteria invalid	Dev Lead	Confirmed
A-03	DatumUncodingToString format is fixed: 'Term {name}: Uncoded By {user} at {date}'	TargetObjectDescription validation fails	Dev Lead	Confirm Day 1
A-04	Existing coding behavior (DatumCodingToString) is unchanged by the fix	Regression TCs invalid	Dev Lead	Confirmed
A-05	DB access (read-only SELECT) is available to QA	Cannot execute DB validation steps	DBA Team	Pending
A-06	Bulk uncode feature is accessible in this build	TC-007 and TC-014 marked N/A if absent	Dev Lead	Confirm Day 1
A-07	Test environment is not shared with concurrent dev deployments	Environment stability risk	DevOps	Confirmed